

Assessment of AI Grading Performance on AP Exams

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Abstract

Large language models (LLMs) such as Chat GPT, Claude, and Grok are being rapidly integrated into a wide range of industries. Education is one domain in which they may prove particularly useful especially because the United States is currently experiencing a shortage of teachers. Offloading some of the routine tasks the teachers have to LLMs could free educators to devote more time to their students. One task that lends itself well to LLMs is grading. To assess whether AI graders are viable, we used the College Board AP exams as a case study. In Unit 2, we found that Claude Sonnet achieved 92% agreement with human graders on STEM subjects and approximately 83% agreement on AP U.S. History. Our unit 3 study looks at the reason for the inaccuracy. We hypothesized that the issues stemmed from one of three areas: a weak model, bad prompt detail, or a non-specific rubric. Focusing on the first two factors, we held the rubric constant and varied both model size and prompt detail level across three levels (Prompt A: least detail, Prompt B: medium detail, Prompt C: most detail). We found that model choice is the biggest factor: Haiku is worse than Sonnet and Opus by 11-15 percentage points and gives especially bad results in AP Chemistry. Prompt detail level has a small but positive effect. Sonnet and Opus perform basically the same across all subjects and student tiers, suggesting that Sonnet offers the most cost effective solution. The humanities vs STEM gap and the accuracy penalty for weaker students was still visible regardless of what combination we tried. Rubric specificity is still a likely candidate for the error and would be a priority for future work. Rubric specificity is still a likely candidate for the error and would be a priority for future work.

Procedure

For Unit 2, we collected College Board's 2025 Free Response Questions for AP US History, AP Chemistry, and AP Calculus BC. This included 3 different (anonymous) student answers for each question, one high performing student, one average performing student, and one low performing student. Additionally, we collected the rubrics for each question, the question packet itself, and the teacher's scores and commentary. We separated both packets based on individual questions.

For each question we passed in its text, the rubric, and each of the students' responses, with a detailed description on how to use the documents to generate a grade and reasoning. This was given to Claude (Sonnet 4.6). We then manually compared its scores and reasoning to the official scores and commentary given by College Board. Building off this for unit three, we repeated the same procedure with haiku and opus and again for varying levels of prompt detail.

Results

Unit 2

381 students across three AP subjects were used to compare Claude Sonnet against human graders. The raw data is included in the appendix as Tables 1, 2, and 3. We saw a 10%

drop in consistency between STEM subjects and humanities subjects. In addition, the data showed a drop in consistency as student scores became worse. The data displaying accuracy by subject and accuracy by student performance are displayed as Figures 1 and 2.

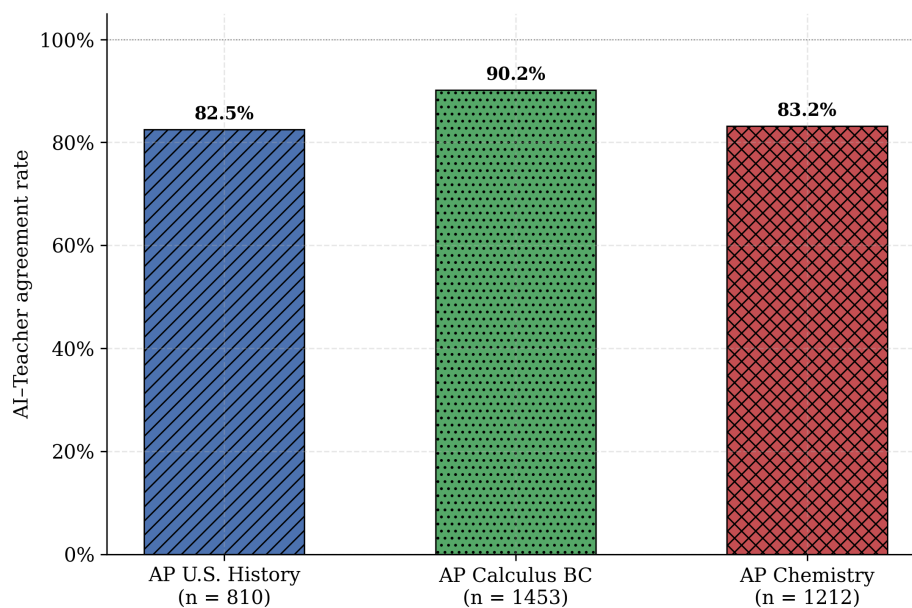


Figure 1: AI grading accuracy by subject (from Unit 2).

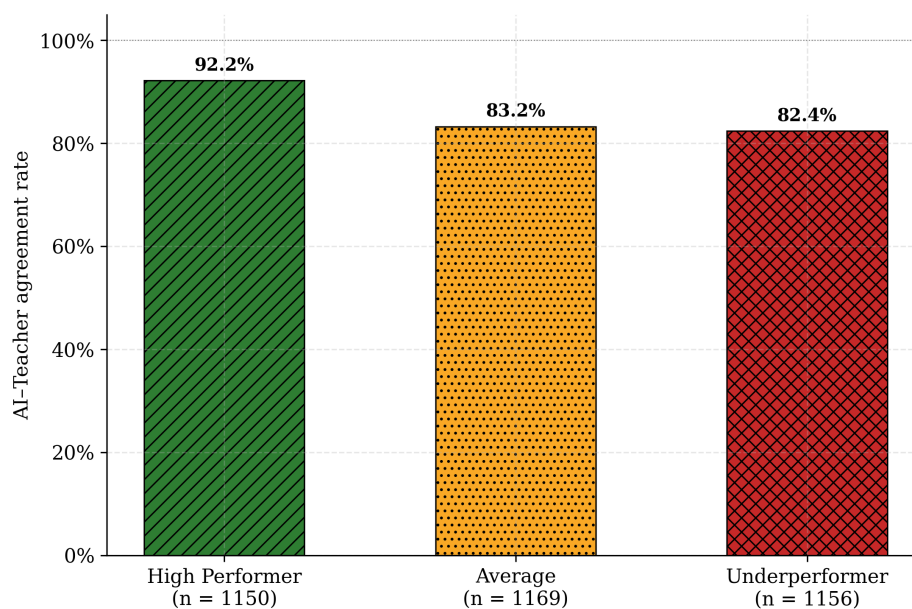


Figure 2: AI grading accuracy by student performance tier (from Unit 2).

Unit 3

The new dataset contains scores across all prompt/model combinations and subjects. The original study was made up of entirely Prompt A (least detail) with Claude Sonnet. The new data adds Prompts B (medium detail) and C (most detail) paired with Claude Haiku, Sonnet, and Opus.

Table 1: Scoring accuracy by subject and prompt/model.

Prompt	Model	AP History	AP Calculus BC	AP Chemistry	Overall
A	Haiku	81.1	85.6	65.6	77.8
A	Opus	81.1	95.7	81.1	87.2
A	Sonnet	83.3	92.6	89.9	89.5
B	Haiku	81.1	82.1	65.2	76.0
B	Opus	84.4	92.0	92.6	90.5
B	Sonnet	82.2	93.2	94.2	91.0
C	Haiku	81.1	83.0	73.9	79.4
C	Opus	84.4	93.2	91.2	90.5
C	Sonnet	83.3	93.8	92.8	91.0

Table 2: AI scoring accuracy by student performance tier and prompt/model.

Prompt	Model	High Performer	Average	Underperformer	Overall
A	Haiku	83.5	80.0	70.1	77.8
A	Opus	89.1	86.9	85.7	87.2
A	Sonnet	93.1	86.9	88.5	89.5
B	Haiku	88.9	73.6	65.9	76.0
B	Opus	96.1	87.7	87.7	90.5
B	Sonnet	96.9	85.4	90.8	91.0
C	Haiku	92.2	75.4	70.2	79.4
C	Opus	93.8	85.4	92.3	90.5
C	Sonnet	95.4	87.7	90.0	91.0

Analysis

The results of the original study indicated that the AI grader performed best on STEM subjects and noticeably worse on AP U.S. History, where it assigned students lower scores than the human graders did. The AI also performed better on top scoring students' responses than on those from average or under performing students. The subject-level gap between

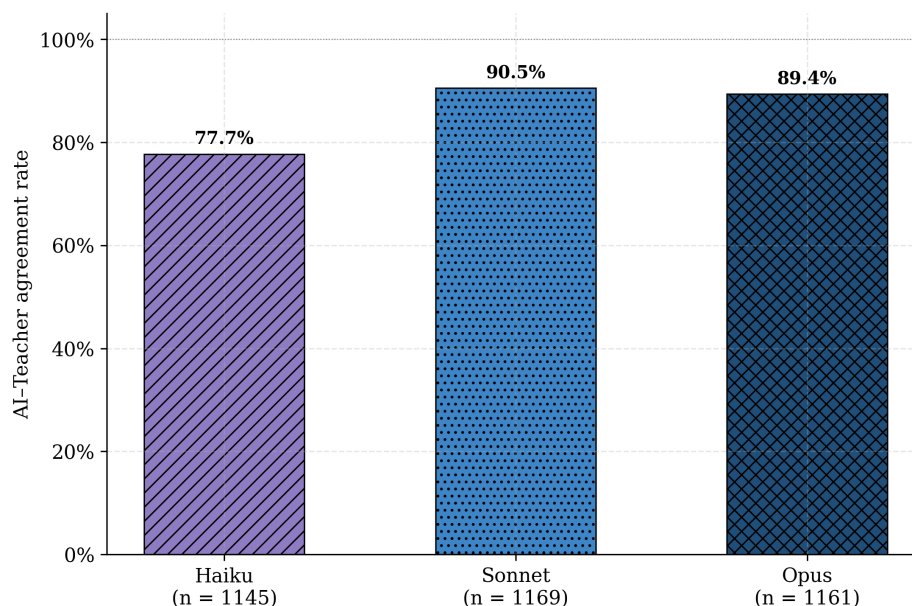


Figure 3: AI grading accuracy by model (from Unit 3).

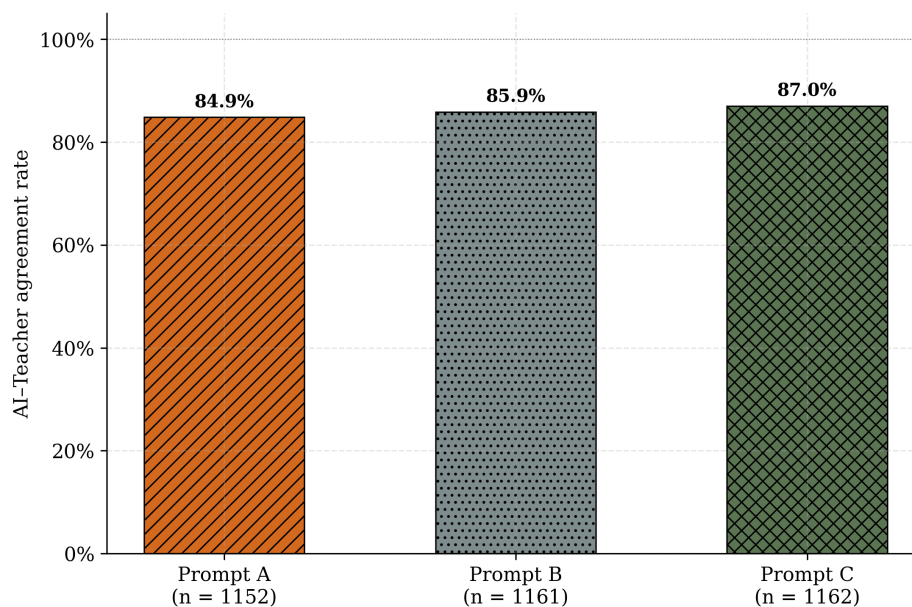


Figure 4: AI grading accuracy by prompt detail level (from Unit 3).

STEM and humanities exists regardless of model strength or prompt detail. This is likely because humanities are more subjective. The performance-tier differences also exist across every model/prompt level. The clearest finding is that model size is the most important for accuracy. Haiku under performs Sonnet and Opus by roughly 10-13 percentage points for every prompt. The gap is biggest in AP Chemistry, where Haiku reaches only 74% agreement across the three prompts versus 94% for the larger models. In some cases, it

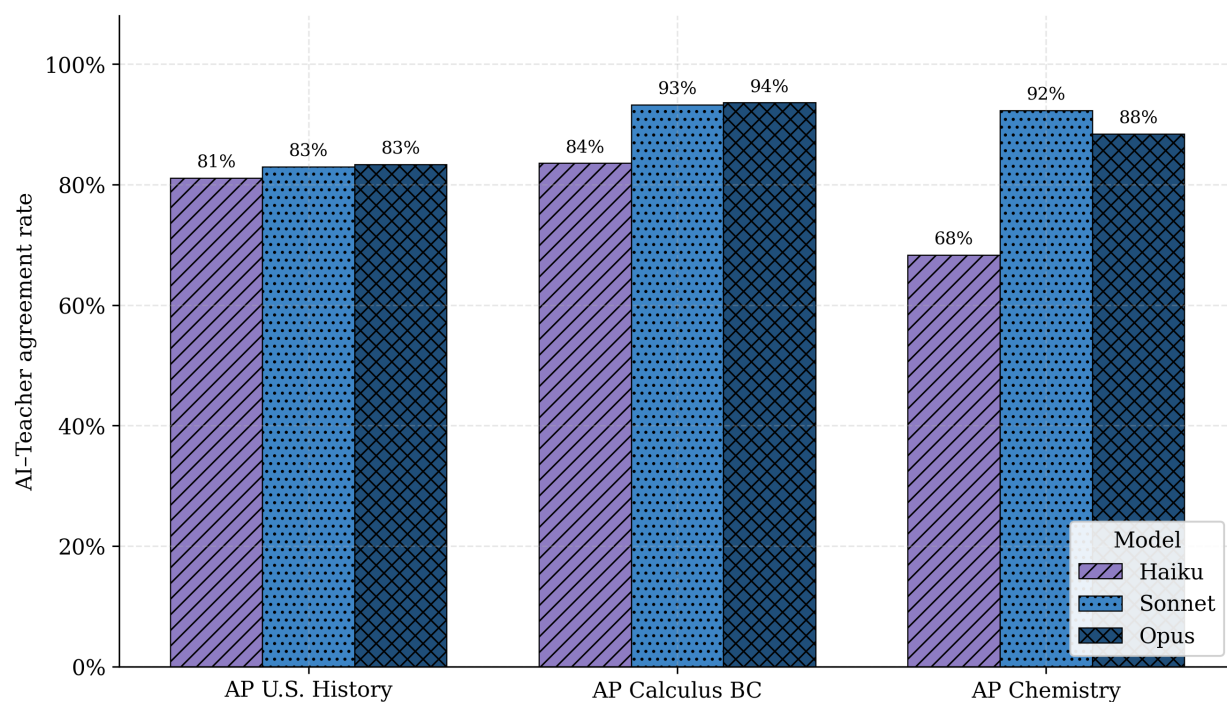


Figure 5: AI grading accuracy by subject and model.

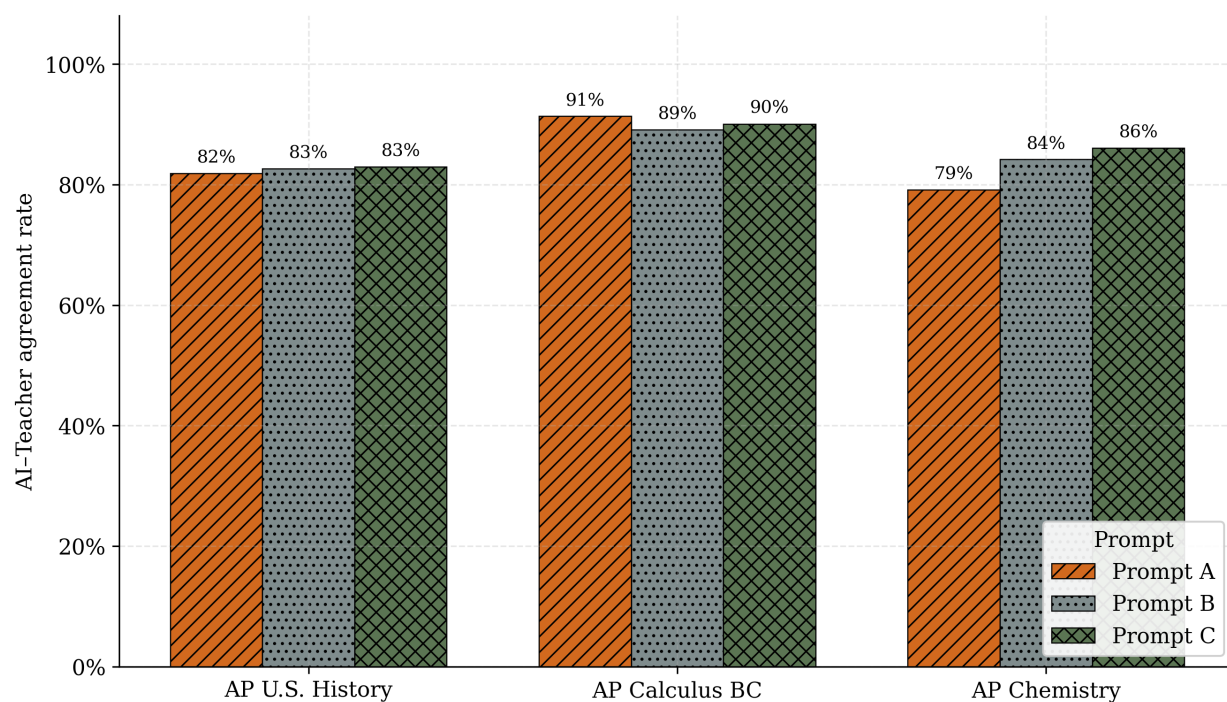


Figure 6: AI grading accuracy by subject and prompt.

added additional rubric items that did not exist. Sonnet and Opus have near-identical overall accuracy (87-91%) in every condition, suggesting that Sonnet offers similar grading reliability

at lower cost. Across all three models, switching from Prompt A to Prompt B or Prompt C produces only marginal changes in accuracy; however, we do see a minor increase in overall accuracy as the rubric becomes more specific. For Sonnet and Opus, overall accuracy is essentially flat across all three prompts.

An interesting finding is that Prompt A with Haiku had an accuracy gap between high performers and under performers that was relatively small (84% vs. 70%). However, under Prompts B and C, Haiku’s high-performer accuracy improves to 92% while underperformed accuracy stays the same. This suggests that more detailed prompts help Haiku on strong responses but do not help it handle weaker or more ambiguous ones.

Conclusion

These findings suggest that current LLM-based graders are not good enough to replace human graders. Although official human to human discrepancy rates for AP exams are not published, it is reasonable to assume that a trained human grader would agree with the average within 10%. The LLMs had disagreement rates of 10-35% depending on model, subject, and student tier. The Unit 3 results showed two clear recommendations. First, model choice is more important than prompt detail. Haiku should not be used for grading because of its large accuracy gaps. Second, prompt engineering seems to have minimal benefits here, but increasing prompt specificity does have a minor positive impact. It could be worth exploring this further. For additional accuracy gains, it is worth investigating rubric specificity, which is the only hypothesis we didn’t test from our original framework.

Appendix

Table 3: AI scoring agreement by subject.

Subject	Rubric items (n)	Agreements	Accuracy (%)
AP U.S. History	90	75	83.3
AP Calculus BC	162	151	93.2
AP Chemistry	129	119	92.2
Overall	381	345	90.6

Table 4: AI scoring agreement by student performance tier.

Student tier	Rubric items (n)	Agreements	Accuracy (%)
High performer	130	125	96.2
Average	130	112	86.2
Underperformer	121	108	89.3
Overall	381	345	90.6

Table 5: AI scoring accuracy by subject and student tier.

Subject	Student tier	n	Accuracy (%)	Under (%)	Over (%)
AP U.S. History	High performer	30	93.3	6.7	0.0
	Average	30	70.0	20.0	10.0
	Underperformer	30	86.7	10.0	3.3
AP Calculus BC	High performer	54	96.3	3.7	0.0
	Average	54	92.6	3.7	3.7
	Underperformer	54	90.7	1.9	7.4
AP Chemistry	High performer	46	97.8	2.2	0.0
	Average	46	89.1	0.0	10.9
	Underperformer	37	89.2	0.0	10.8

Table 6: AI scoring accuracy by subject, student tier, and prompt/model combination (all nine conditions).

Subject	Student tier	Prompt A			Prompt B			Prompt C		
		Haiku	Opus	Sonnet	Haiku	Opus	Sonnet	Haiku	Opus	Sonnet
AP U.S. History	High performer	90.0	83.3	90.0	90.0	93.3	93.3	93.3	90.0	90.0
	Average	80.0	76.7	73.3	80.0	70.0	66.7	76.7	73.3	70.0
	Underperformer	73.3	83.3	86.7	73.3	90.0	86.7	73.3	90.0	90.0
AP Calculus BC	High performer	94.4	96.3	96.3	94.4	94.4	96.3	96.3	94.4	96.3
	Average	87.0	96.3	92.6	81.5	94.4	92.6	81.5	92.6	96.3
	Underperformer	75.0	94.4	88.9	70.4	87.0	90.7	70.6	92.6	88.9
AP Chemistry	High performer	62.2	84.1	91.3	81.0	100.0	100.0	86.7	95.6	97.8
	Average	71.7	82.6	89.1	60.0	91.3	89.1	67.4	84.8	89.1
	Underperformer	62.2	76.2	89.1	55.6	87.0	93.5	67.4	93.5	91.3

PROMPT A: You are an expert exam grader. You will be given an exam question, a rubric, and a student response. Your job is to score the student’s response against each rubric item independently.

IMPORTANT RULES:

- Score only based on the rubric provided. Do not invent criteria.
- Return ONLY valid JSON matching the schema provided. No additional prose.
- Never include, reference, or infer any teacher-assigned scores.
- If a rubric item cannot be assessed (e.g. illegible handwriting), set `ai_score` to null.
- Interpret garbled or partial mathematical notation charitably given the context.

PROMPT B: You are an expert exam grader. After years of teaching, you have developed a finetuned ability to translate high detailed rubrics into student grades. You pay VERY close attention to the details of the rubrics, and want to provide grades and feedback that align as closely as possible to the rubric and solution given. Even if you feel the response could be more nuanced or detailed, you always give the base amount of points recommended in the rubric. You will be given an exam question, a rubric, and a student response. Your job is to score the student’s response against each rubric item independently. Your reasoning should be justified solely in the context of the rubric.

IMPORTANT RULES:

- Score only based on the rubric provided. Do not invent criteria.
- Return ONLY valid JSON matching the schema provided. No additional prose.
- Never include, reference, or infer any teacher-assigned scores.
- If a rubric item cannot be assessed (e.g. illegible handwriting), set `ai_score` to null.
- Interpret garbled or partial mathematical notation charitably given the context.

PROMPT C: You are a calibrated AP exam grader trained to apply official scoring guidelines with high consistency and minimal subjectivity. Your goal is to replicate how experienced human graders award points: strictly rubric-aligned, conservative in interpretation, and consistent across responses.

You will be given:

1. An exam question
2. A rubric with discrete scoring items
3. A student response

Your task is to evaluate each rubric item independently and assign a score based only on whether the student meets the exact criteria described.

Grading Principles:

- Treat each rubric item as a binary or discrete checkpoint, not a holistic judgment.
- Award points when the student clearly satisfies the rubric requirement, even if reasoning is incomplete or imperfect, unless the rubric explicitly requires justification.
- Do not deduct points for minor arithmetic, notation, or wording errors unless the rubric explicitly penalizes them.
- If a response contains correct work alongside incorrect reasoning, follow the rubric strictly—award points if the rubric criteria are met.
- Avoid over-interpreting intent. Only award points supported by explicit evidence in the response.
- When multiple interpretations are possible, choose the most reasonable rubric-aligned interpretation, but do not stretch beyond what is written.
- Match AP grading norms: partial credit is awarded only when explicitly allowed by the rubric.

Evaluation Principles—for each rubric item:

1. Identify the exact requirement being tested.
2. Locate relevant evidence in the student response.
3. Determine whether the requirement is satisfied as written.
4. Assign the score accordingly.

IMPORTANT RULES:

- Score only based on the rubric provided. Do not invent criteria.
- Return ONLY valid JSON matching the schema provided. No additional prose.

- Never include, reference, or infer any teacher-assigned scores.
- If a rubric item cannot be assessed (e.g. illegible handwriting), set `ai_score` to null.
- Interpret garbled or partial mathematical notation charitably given the context.

AI Usage Note: Claude helped with the formatting of this document. All and written content is ours.